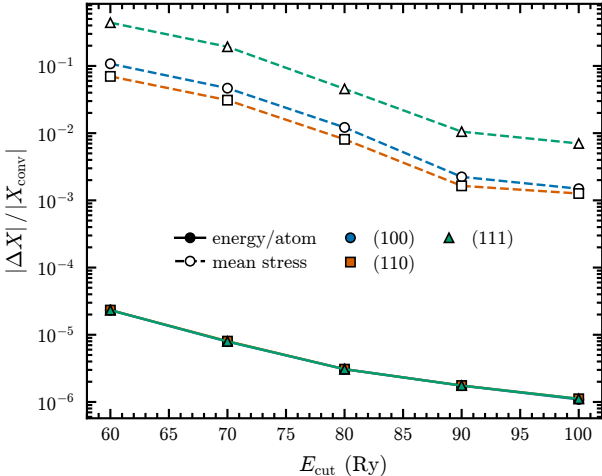




Why converging the total energy does not converge the stress. Both quantities are plotted as $|\Delta X|/|X_{\text{conv}}|$ relative to their own value at 120 Ry, each divided by its own converged magnitude, so the two are dimensionless and directly comparable on one log axis; the y value is the fraction of itself that each quantity is still moving by. At 60 Ry the energy per atom is already converged to 2×10^{-5} while the mean in-plane stress is only at 4×10^{-1} , and at 100 Ry they stand at 1×10^{-6} and 7×10^{-3} . At every cutoff sampled the stress is between 935 and 24227 times less converged than the energy, because the energy is variational in the basis-set error while the stress, its first derivative, is not. A cutoff chosen on an energy criterion alone is therefore not a converged setting for a stress campaign -- the lesson that set 90 Ry here, and the one that recurred on every other axis of this study.